

MARTÍN HERNÁNDEZ SALINAS

Postdoctoral researcher at the UCLA Department of Statistics and Data Science. My research focuses on control theory, partial differential equations, randomized numerical methods, and the theoretical foundations of AI, particularly in deep learning, and reinforcement learning.

CONTACT

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-  @Martinshs
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SCIENTIFIC COMPUTING

-  Development of numerical methods and Python implementations for optimal control problems governed by partial differential equations.
-  Implementation of randomized algorithms for parabolic PDEs, dynamical systems, and graph-based models.
-  Design and implementation of deep-learning and reinforcement-learning methods, including continuous-time and mean-field models.
-  Experience with Python, MATLAB, FEniCS/DOLFIN, PyTorch, TensorFlow, NumPy, SciPy, and Matplotlib.

Selected implementations are available on GitHub:  @Martinshs

Languages

- Spanish
- English
- Portuguese
- German



RESEARCH INTERESTS

- Control Theory
- Partial Differential Equations
- Numerical Analysis
- Deep learning
- Machine learning
- Efficient algorithms
- Optimization

EDUCATION

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|---|-------------------------------------|------------------------------|
|  11/2025 - | UCLA Statistics & Data Science, USA | Postdoctoral Researcher |
|  09/2021 - 09/2025 | FAU Department of Maths, Germany | Dr. rer. nat. in Mathematics |
|  03/2019 - 03/2021 | UTFSM Department of Maths, Chile | M.Sc. in Mathematics |
|  03/2015 - 03/2021 | UTFSM Department of Maths, Chile | Mathematical Engineer |

HONORS & AWARDS

-  Ph.D. degree awarded summa cum laude, the highest distinction in the German academic system.
-  M.S. degree with the highest distinction.
-  Mathematical Engineering degree with the highest distinction.
-  (2016 - 2021) Honor roll member of the Technical University Federico Santa María. Student belonged to the USM Honor Roll, whose members, by academic performance, are located within the top 10% of students taking subjects

TEACHING EXPERIENCE

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|  03/2018 - 12/2020 | UTFSM, Chile | Exercise Teacher |
| Exercise teacher for eight courses related to differential and integral calculus, and ordinary and partial differential equations. | | |
|  03/2021 - 12/2021 | UTFSM, Chile | Part-Time Teacher |
| Lecturing at the Mathematics Department. | | |
|  05/2024 | ML&OC24, Italy | Lecturer at Summer School |
| Summer school in the ML&OC 24: Machine Learning and Optimal Control. | | |
|  06/2024 | EECI-IGSC M18, Croatia | Lecturer at Summer School |
| Summer school in EECI-IGSC 2024: Control and Machine Learning. | | |
|  01/2025 | UFF, Brazil | Lecturer at Summer School |
| Summer school at UFF, Introduction to turnpike phenomenon | | |

PROFESSIONAL EXPERIENCE

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|  12/2022 - 09/2025 | TRR154 subproject C07, Germany. | Associated Ph.D. student |
| Analysis of gas transport models, numerical methods and implementations. | | |
|  02/2021 - 02/2022 | Deloitte Spa. Santiago | Analyst in Financial Risk |
| Specialist in financial risk models, statistical models and programming. | | |

FUNDING

2019-2021	Scholarship. Dirección de Postgrados y Programas (DPP) of the U. Técnica Federico Santa María.
2021-2022	Scholarship. Deutsche Forschungsgemeinschaft within Project ID 239904186 – TRR 154 Mathematical modeling, simulation and optimization using the example of gas networks.
2022-2026	Scholarship. Acuerdo Bilateral en el Extranjero ANID-DAAD.
2022-2026	SFB Transregio 154 Mathematische Modellierung, Simulation und Optimierung am Beispiel von Gasnetzwerken.
2023-2025	DAAD/CAPEs Programs for Project-Related Personal, grant 57703041, Name: 'Control and numerical analysis of complex systems'.

TALKS AND ACTIVITIES

● 07/2026	Talk at the conference Dynamics, Deep Learning, and Control in Nonlinear PDEs, in the Ettore Majorana Foundation and Centre for Scientific Culture. Erice, Italy.
● 05/2026	Talk at the conference Multi-Agent and Mean Field Reinforcement Learning, Chicago, USA.
● 04/2026	Talk at the Southern California Applied Mathematics Symposium, Los Angeles, USA.
● 01/2025	Talk at Technical University of Darmstadt and visit to Prof. Jan Giesselmann.
● 01/2025	Lectures at Universidad Federal Fluminense, Brazil, and visiting Prof. Juan Límaco.
● 12/2024	Talk at Humboldt University of Berlin, and visit to Prof. Falk Hante.
● 11/2024	Attendance at Oberwolfach Seminar: Control and Machine Learning, Germany.
● 26/08/2024	Talk at X Partial differential equations, optimal design and numerics, Benasque, Spain.
● 23/08/2024	Talk at X Partial differential equations, optimal design and numerics, Benasque, Spain.
● 19/08/2024	Talk at X Partial differential equations, optimal design and numerics, Benasque, Spain.
● 06/2024	Lecturer at EEIC-IGSC 2024: Control and Machine Learning (M18), Dubrovnik, Croatia.
● 05/2024	Lecturer at the ML&OC24: Machine Learning and Optimal Control summer school, Gaeta, Italy.
● 11/2023	Visit to Professor M. Lazar for a week at the University of Dubrovnik, Croatia.
● 10/2023	Visit to the University of Deusto in Spain together with Prof. Falk Hante from the Humboldt-Universität zu Berlin.
● 08/2023	Talk at the 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo.
● 02/06/2023	Talk at the 93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM), Germany.
● 01/06/2023	Talk at the 93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM), Germany.
● 05/2023	Talk at the "Orientierungswoche" for first-year students of the Data Science Department at FAU, Germany.
● 11/2022	Attendance at the workshop in Good Scientific Practice, Bamberg, Germany.
● 09/2022	Talk at the 25th International Symposium on Mathematical Theory of Networks and Systems, Germany.
● 08/2022	Talk at the IX Partial differential equations, optimal design, and numerics, Spain.
● 08/2022	Talk at the IX Partial differential equations, optimal design, and numerics, Spain.
● 04/2022	Talk at South Zone Mathematics Conference, Chile.
● 10/2020	Doctoral school, of the Universidad Catolica, Chile.
● 2019-2020	Member of the student council of Sciences, an organization that groups five careers of the Technical University Federico Santa María, which is in charge of the student representation.
● 01-03/2018	Student at summer school at Instituto de Matematica Pura e Aplicada (IMPA).

PUBLICATIONS

Averaged Turnpike Property for Differential Equations with Random Constant Coefficients

👤 **Martín Hernández, Rodrigo Lecaros, and Sebastián Zamorano**

📅 2023 📖 *Mathematical Control and Related Fields*, **13**(2), 808–832

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DOI:10.3934/
mcrf.2022016

Uniform Turnpike Property and Singular Limits

👤 **Martín Hernández and Enrique Zuazua**

📅 2024 📖 *Acta Applicandae Mathematicae*, **190**, Paper No. 3, 33 pp.

🔗
DOI:10.1007/
s10440-024-
00640-7

Mini-batch Descent in Semiflows

👤 **Alberto Domínguez Corella and Martín Hernández**

📅 2025 📖 *ESAIM: Control, Optimisation and Calculus of Variations*, **31**, Article 28

🔗
DOI:10.1051/
cocv/2025018

Turnpike phenomenon for averaged optimal control

👤 **Martín Hernández, Martín Lazar, Sebastián Zamorano.**

📅 2026 📖 *Optimization*, 1–24.

🔗
DOI:10.1080/
02331934.2026
.2615776

PREPRINTS

Constructive Universal Approximation and Finite Sample Memorization by Narrow Deep ReLU Networks.

👤 **Martín Hernández and Enrique Zuazua.**

📅 2025 📖 Preprint

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arXiv:2409.06555

Random Batch Methods for PDE control on graphs.

👤 **Martín Hernández and Enrique Zuazua.**

📅 2025 📖 Preprint

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arXiv:2506.11809

Random domain decomposition for parabolic PDEs.

👤 **Martín Hernández.**

📅 2025 📖 Preprint

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arXiv:2508.21557

Convergence, design and training of continuous-time dropout as a random batch method.

👤 **Antonio Álvarez-López and Martín Hernández.**

📅 2025 📖 Preprint

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arXiv:2510.13134

Policy Gradient for Continuous-Time Mean-Field Control.

👤 **Erhan Bayraktar, Martín Hernandez, Qinxin Yan, Yuhua Zhu.**

📅 2026 📖 Preprint

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arXiv:2605.20718

Mean-Field PhiBE: Continuous-Time Mean-Field Reinforcement Learning from Discrete-Time Data.

👤 **Erhan Bayraktar, Martín Hernandez, Qinxin Yan, Yuhua Zhu.**

📅 2026 📖 Preprint

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arXiv:2606.26498